BICYCLE OPTIMIZATION PROJECT – STAGE 2 – ME303

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Manufacturer: Huffy

Model: Rock Creek 26”

Contents

Overview

This project investigates the static failure behavior of a bicycle frame and fork system using computer aided design (CAD) modeling and engineering stress analysis. A simplified 3D model was created in SOLIDWORKS with idealized circular tube geometries. Distortion Energy Theory was applied to evaluate yielding in the frame, identifying the seat tube as the critical member. Static failure occurred at a rider weight of 1915 lbf, demonstrating a high factor of safety (>9) under normal loading. The fork-steerer interface was analyzed under combined bending and torsional loading using Maximum Shear Stress theory, resulting in a factor of safety of 23.39. An impact-based impulse–momentum analysis predicted failure from yield for a vertical drop height of 6.35 ft. To increase height limit by 50%, a redesigned fork diameter of 1.60 in was selected. While these results demonstrate a generous safety margin under expected conditions, they represent idealized static loading only. Real-world conditions such as fatigue, tire compression and suspension damping were not included. Thus, the analysis provides a conservative estimate of structural performance, suitable for evaluating extreme single use loads. These methods also demonstrate how minor geometric changes can significantly enhance performance.

CAD Model

Methods

The steering column, which includes the fork, and the handlebars were added to the CAD bike frame model from Stage 1 of this project. This was completed using reference geometry planes and extruding circular cross-sections with the Boss/Bose and Sweep features in SOLIDWORKS.

<u>Assumptions</u>

- **All components are circular tubes**. This simplifies the model and is accurate to typical bike geometry.
- **Grips and gear system were removed from the handlebars for simplicity.** Removal of these features has no impact on the static analysis of the frame and fork.
- **All steering component tubing is solid.** This simplifies the 3D printing of the frame and steering components and mirrors real-world components for strength.

Results

The following figure 1 shows a comparison of the CAD model of the bicycle frame and steering components to a real photo of the bicycle.

Figure 1:

Isometric Profile Comparison of CAD Model and Bicycle Photo

Static Failure - Frame

Methods

To determine the static failure of the bicycle frame, the Distortion Energy (DE) theory was selected as the appropriate ductile failure criteria. Carbon steel (1020) has an elongation of approximately 15% (Axiom, 2013), which exceeds the 5% threshold used to distinguish ductile from brittle materials. Compared to the Maximum Shear Stress (MSS) theory, DE provides a more comprehensive assessment of yielding under uniaxial loading, accounting for the combined effect of all three principal stresses. In contrast, MSS only considers maximum shear stress.

Using the Stage 1 spreadsheet, a maximum allowable force, the effective stress and the factor of safety for each frame member is calculated using the frame material's yield strength, the axial stresses from the internal reaction forces and cross-sectional area. The effective stress simplifies to axial stress as there is only uniaxial loading, thus there will only be a single principal stress (either σ_1 in tension or σ_3 in compression) and no shear. The applied rider weight is then scaled in upward increments until the axial stress in one of the members reaches yield (a factor of safety of one), which determines the maximum weight the frame can support before static failure.

Assumptions

- **The applied load is modeled as a vertical point load at the seat location, representing the full rider weight.** This approximates real loading conditions under static seated posture, with weight centered over the saddle.
- **All members are assumed to have uniform cross-sectional areas and thin-walled circular tubing.** Some bicycles have butted tubes (wall thickness is greater at the ends and thinner in the middle). Using a uniform thickness will simplify calculations and still be consistent with entry grade bicycles such as the model in this study.
- **Distortion Energy Theory (Von Mises Theory) is used as the failure criteria.** This assumes failure occurs when the effective stress reaches the critical value corresponding to the material's yield strength. Since the frame is composed of a ductile material (elongation > 5%) and the members are predominantly subjected to uniaxial stress, the effective stress simplifies to the axial stress.
- **Neglect bending, shear and torsion due to experiencing a sole vertical point load from the rider's weight.** The frame geometry resembles a truss, and members are oriented such that axial loading dominates under static loading. This simplifies static failure analysis as Distortion Energy (Von Mises) effective stress (σ') will simplify to axial stress under uniaxial loading.

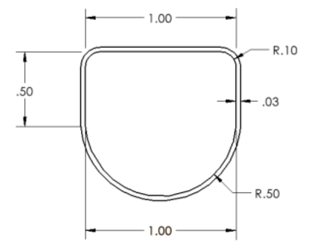
Cross-Sections

Cross-sectional areas of each frame tube are shown in the following Tables 1 and 2 and summarized in Table 3 on page 4. Wall thickness for all tubes is 0.03 inches, except for the seat tube wall thickness, which is 0.04 inches. The seat stays and the chain stays have the same cross-sections and cross-sectional area.

All images are scaled 2:1 for readability.

Table 1.

Cross-sectional diagrams of bicycle frame top tube, seat tube and chain/seat stays.

<table>
<tr><th>Top Tube</th><th>Seat Tube</th><th>Seat Stays/Chain Stays</th></tr>
<tr>
<td>
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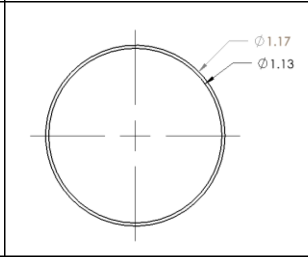
<td>
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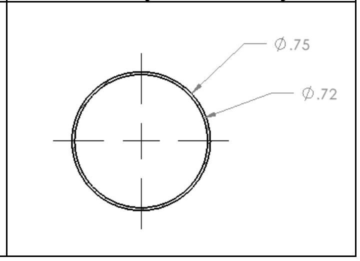
<td>
</td>
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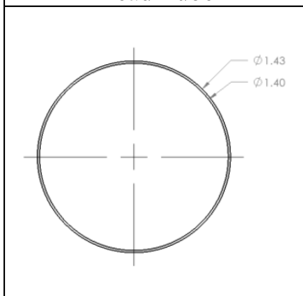
</table>

Table 2.

Cross-sectional diagram of bicycle frame head tube and down tube.

<table>
<tr><th>Head Tube</th><th>Down Tube</th></tr>
<tr>
<td>
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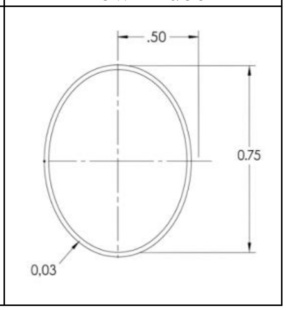
<td>
</td>
</tr>
</table>

Table 3.

Summarized cross-sectional dimensions and area for Each frame tube.

Tube Description	Inner Diameter (in)	Outer Diameter (in)	Cross-Sectional Area (in^2)
Top Tube	0.5	0.53	0.12
Seat Tube	1.13	1.17	0.07
Seat Stays	0.72	0.75	0.03
Head Tube	1.40	1.43	0.07
Down Tube	0.5	0.53	0.12
Chain Stays	0.72	0.75	0.03

Results

Using the static analysis spreadsheet created in Stage 1, the maximum allowable force (σ_{max}), the effective stress (σ') and the factors of safety for each frame member was evaluated using distortion energy with results for factor of safety ranging from 9 to 49. Calculations are shown in Figure 2 on the next page.

Figure 2.

Static Analysis Spreadsheet Calculation for 200lbf Load

Static Failure Analysis			
Yield Strength (σy)(psi)	50800		
Max Allowable Stress (σmax) (lbf)		Effective Stress (σ') (psi)	FOS (σmax/σ')
Down Tube	6096	2475	20.52891
Top Tube	5588	-2328	21.81950
Seat Tube	3556	5306	9.57491
Seat Stays	2540	-1024	49.62479
Chain Stays	1524	-3431	14.80588
**Note: Based on cross-sectional area*			**Note: FOS = 1 = Yield*

The rider weight was then incrementally increased by 100lbf until the factor of safety for one of the members dropped below 1—indicating the frame yielded. The seat tube reached yield first and corresponded to a maximum rider weight of 1915lbf. The results of the evaluation of the maximum rider weight are shown in Figure 3 below.

Figure 3:

Static Analysis Spreadsheet Calculation for Critical Load (1915 lbf)

Input Values	
Bicyclist Weight (W) [lbf]	**1915**
Angles (°)	
<BAD	34
<ABD	56
<DBC	68
<BCD	55
<CDB	57
AB from horizontal	34
BC from horizontal	6
AD from horizontal	16
BD from horizontal	74
Headtube	16
Member Lengths (in)	
Length AD	20
Length AB	24
Length BC	17
Length BD	15
Length CD	18
Lever Arms	
ax	19.897
bx	4.135
cx	16.907
dx	19.225
Length A to C (ax + cx)	36.804
Cross-Sectional Area	
Top Tube	0.11
Seat Tube	0.07
Seat Stays	0.05
Head Tube	0.07
Down Tube	0.12
Chain Stays	0.03

External Reaction Forces (lbf)	
Reaction at A (FA)	914.659
Reaction at C (FC)	500.170

**Note 2 FC reactions (2 identical rear supports)*

Tension = Postive

Compression = Negative

Check:	
Total Reaction (FA + 2FC)	1915

Static Analysis

Internal Reaction Forces (lbf)	
FAB (Down Tube)	2843.268
FAD (Top Tube)	-2452.169
FBD (Seat Tube)	3556.033
FDC (Seat Stays)	-490.088
FBC (Chain Stays)	-985.575

Static Failure Analysis			
Yield Strength (σy)(psi)	50800		
Max Allowable Stress (σmax) (lbf)		Effective Stress (σ') (psi)	FOS (σmax/σ')
Down Tube	6096	23694	2.14401
Top Tube	5588	-22292	2.27880
Seat Tube	3556	50800	0.99999
Seat Stays	2540	-9802	5.18275
Chain Stays	1524	-32852	1.54631
**Note: Based on cross-sectional area*			**Note: FOS = 1 = Yield*

Critical Bicyclist Weight (lbf)	**1915**

Interpretation

Static analysis of the bicycle frame identified the seat tube as the critical member, with yielding at a rider weight of 1915 lbf. The seat tube being the member to yield first is a reasonable outcome, as it directly supports the rider weight and is oriented at a steep angle. making it highly loaded in axial compression.

The calculated failure load far exceeds that of an average adult (150–200 lbf), indicating a significant safety margin under normal static condition and is demonstrated by the high factor of safety (9.57) under a 200lbf load as shown in Figure 2. This margin seems appropriate, given that the analysis does not account for multi-directional forces that may be applicable during real-world riding. This includes multi-directional factors from internal sources in the bicycle and from external factors such as riding in rough terrain where vertical drops and or other obstacles that could cause stress on the frame may be encountered.

Static Failure - Fork

Methods

To determine the static failure at the fork blade-steerer interface, the reaction force F_A that acts vertically on the headtube (calculated in Stage 1 for this project under a 200lbf load) is split equally between the fork blades and split into individual components due to the angle of the fork. Using a simplified free body diagram, stress analysis is completed and the critical stress element for the interface is drawn. Factor of safety is then evaluated both analytically using the Maximum Stress theory ($FOS = \frac{\sigma_y}{\sigma 1 - \sigma 3}$) and graphically using a MSS stress envelope and load lines.

The load that will cause yielding in the system is then evaluated using stress analysis and distortion energy principles. Impulse-momentum principes are then applied using this yield force to identify the vertical drop distance that will cause static failure over a 0.05 second impact time. Finally, design optimization is performed to increase the allowable drop height by 50%. This is achieved by using the same impulse equation to determine the required impact force from the increased height and performing stress analysis in terms of an unknown tube radius. The distortion energy theory is again applied, setting the factor of safety to one to solve for the minimum radius needed to prevent yielding. The final optimal diameter is selected from standard sizes in the textbook by rounding up to the nearest available option.

Assumptions

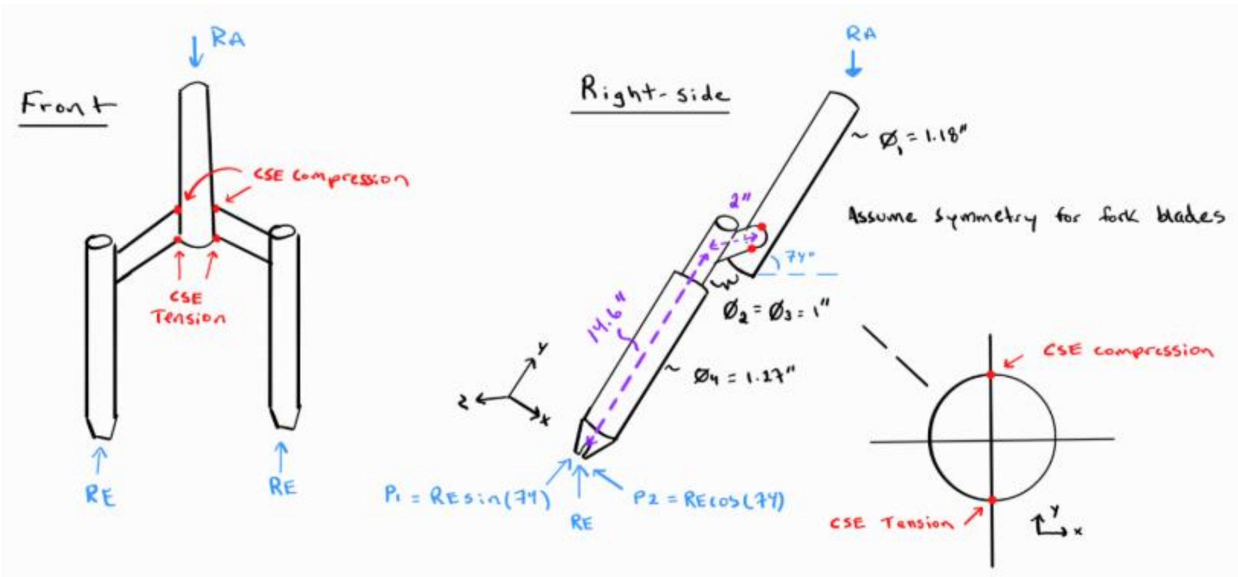
- **All members (fork blade, steerer) are assumed to have circular cross sections.** While actual interface shape is slightly elliptical, elliptical shear analysis is beyond the course scope and not covered in the Shigley's textbook. This may slightly underestimate shear stress (due to larger area) and overestimate bending stress (due to lower moment of inertia).
- **Forks are modeled as solid circular sections.** This assumption increases stiffness which aligns with design priorities in mountain bikes where forks must resist high bending loads from impacts and rough terrain.
- **The vertical reaction at the headtube from Stage 1 is used to represent rider load through the forks.** This accounts for the vertical load based off a 200 lbf rider weight, and that the rider does not sit directly over the front wheel.
- **Vertical drop impact is transferred through the frame in one pulse (a static equivalent force).** This simplification is appropriate for calculating the static failure of a single vertical drop as there is no repeated loading. This may not be suitable for fatigue or dynamic analysis which entail repeated loading.
- **No weld or stress concentrations are considered at the blade-steerer interface for simplicity.** This allows stress to be uniformly distributed across the cross-section of the interface. This simplification is consistent with problems completed in coursework; inclusion would require advanced methods outside of the scope of this course.

Critical Stress Element

To locate the critical stress element at the fork blade–steerer interface, loading on a single fork blade was analyzed under symmetry (two identical blades). The total reaction force at the headtube (R_A = 95.526 lbf from the static analysis spreadsheet) was equally divided between the two blades (R_E = 47.763 lbf). The blade reaction force was resolved into its directional components due to the 74° angle of the forks and is shown in a free body diagram of the fork-blade and steerer system in Figure 4.

Figure 4.

Freebody Diagram of Bicycle Fork Blade System and Critical Stress Element



The y-directional force produces a normal bending stress in the two inch member along the z-axis, and the x-directional force produces a torsional shear clockwise around the z-axis at the interface. The combined stresses create a critical stress element with equal tension and compression at the top and bottom of the interface along the outer circumference as shown in figure 4 above.

All supporting calculations may be found in Appendix A and a summary of the forces, moments and stress components may be found in Table 4 below.

Table 4.

Summary of calculated forces, moments and stress components for the fork-steerer system

Force	Value (lbf)
Blade Reaction (R_E)	47.763
Blade Reaction y-component (P_1)	13.165
Blade Reaction x-component (P_2)	45.913
Moments	**Value (lbf-in)**
Bending Moment ($M_{z\ bend}$)	91.826
Torsional Moment ($M_{z\ T}$)	192.209
Stress Components	**Value (psi)**
Bending Stress (σ_{bend})	935.092
Torsional Shear (τ)	979.659

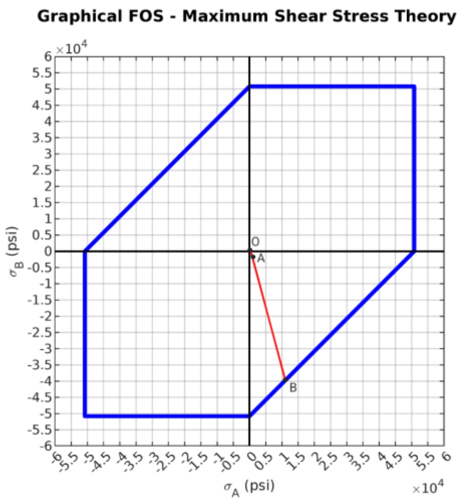
Using the stress components shown in table 4, Maximum Shear Stress theory (MSS) was applied to calculate principal stresses resulting in $\sigma_1 = 1553$ lbf and $\sigma_2 = -618$ lbf and an overall factor of safety of 23.39 for the critical stress element. Calculations for the principal stress and factor of safety may be found in appendix A-x

In conjunction to the calculated analytical factor of safety, a graphical approach was also analyzed. The MSS failure envelope with load lines corresponding to the principal stresses and point of yield is shown in figure 5.

Load line OA was found to be 1/8 inch and load line OB was found to be 14/8 inches (on a 6x6 inch figure scale). This resulted in a graphical factor of safety of 14 (using $FOS = \frac{OB}{OA}$).

Figure 5.

MSS Failure Envelope for the Critical Stress Element



Interpretation – Stress Analysis and Factor of Safety

The stress analysis at the fork blade–steerer interface identified a combined bending and torsional loading condition, producing two equal critical stress elements along the outer circumference of the fork blade (one compression and one in tension). This highlights both the top and bottom of the interface are critical and should be considered when evaluating and optimizing the system.

While both the analytical and graphical methods confirm a largely safe design, the factor of safety values differ—23.39 analytically vs. 14 graphically. This discrepancy likely arises from the limited precision of the graphical method, which uses scaled measurements of load lines. Deviations in line length or scale interpretation can lead to noticeable variation. The analytical method uses the exact stress values which are more reliable for quantitative design decisions, while the graphical approach is best suited for quick visualization of the failure margins for the critical stress elements. Since both factors of safety are significantly above unity it confirms that the fork-steerer interface is well within safe operating limits under an average rider load.

Results – Maximum Drop Distance

To determine the maximum vertical drop height the fork-steerer interface can withstand before static failure due to yielding, a combination of distortion energy theory and impulse–momentum analysis is used. All supporting calculations are provided in Appendix A.

First, the stresses due to bending and torsion at the blade-steerer interface were expressed as functions of an unknown force (F) which would cause yielding. These stress functions were combined into distortion energy effective stress, and the factor of safety was set to one to represent yielding. Solving for the unknown force required to reach the yield limit produced a result of 1252 lbf per fork blade.

Next, the rider's weight was converted to mass resulting in 6.211 slugs ($\frac{lbf}{ft/s^2}$). Using the impulse–momentum equation, the peak impact force from a vertical drop was expressed in terms of height in feet (h) as $F_{impact} =$

$997 \times \sqrt{h}$ .To find the critical drop height, the total yield force, which included two blades (2504 lbf), was set equal to the impact force. Solving for height yielded a maximum drop of 6.35ft (or 76.2 inches) before failure.

Interpretation – Maximum Drop Distance

The predicted maximum drop height of 6.35ft is relatively large compared to common dropping scenarios (dropping off a curb for example, which is less than a foot). This indicates that the fork was overdesigned for durability and safety during normal riding scenarios. A 6ft drop would be more consistent with an extreme drop scenario such as riding over a jump. Additionally, this analysis assumes the entire impact force is transferred instantly through the fork blades, neglecting energy absorption from the suspension system or tire compression, thus the calculated drop height likely underestimates the true failure point, making it a conservative estimate. To increase allowable drop height, a design selection could include optimizing the fork blade radius or selecting higher-strength materials to raise the yield threshold.

Results – Dimensional Change

To increase the allowable drop height by 50% (to approximately 9.53 ft), the same impulse-momentum relationship in the maximum drop distance calculations was applied ($F_{impact} = 997 \times \sqrt{h}$). Substituting the increased height yielded a required impact force of 1539 lbf per fork blade. This force was then reintroduced into the bending and torsional stress equations, with both stress components expressed in terms of an unknown cross-sectional radius.

Following the similar procedure as in the maximum drop analysis, the distortion energy equation was used to combine the stresses into effective stress and the minimum radius required to prevent yielding was evaluated using a factor of safety of 1 and yield strength of 50.8 ksi. The resulting minimum radius was 0.725 inches, corresponding to a diameter of 1.45 inches.

Referencing Table A–17 from *Shigley's Mechanical Engineering Design* on page 1071, a standard diameter of 1.60 inches was selected as the nearest practical size to meet the required strength. All supporting calculations may be found in Appendix A.

Interpretation – Dimensional Ana

The dimensional analysis shows that increasing the allowable drop height by 50% requires only a modest increase in fork/blade diameter. This demonstrates that small geometric changes can significantly enhance performance. However, a larger diameter does have downsides, such as adding weight and increased stiffness, which could affect handling and rider comfort. Reapplying the distortion energy and impulse–momentum methods in reverse highlights a scalable, performance-driven approach that could be extended to other frame components as well or adapted for material optimization in future design iterations.

References

*AISI 1020 carbon steel. (*2013). AZoM.com. https://www.azom.com/article.aspx?ArticleID=9145.

Nisbett, J. Keith. *Shigley's Mechanical Engineering Design*. 12th ed., McGraw-Hill Education, 2024.

Appendix A:

STATIC FAILURE – FORK

$\Sigma F_y = 0 \rightarrow -R_A + 2R_E = 0 \rightarrow R_E = \frac{R_A}{2} = \frac{95.526}{2} = 47.763 \text{ lbf}$

$\Sigma F_x = 0$

$P_1 = R_E \sin 74 = 47.763 \times \sin 74 = 45.913 \text{ lbf}$

$P_2 = R_E \cos 74 = 47.763 \times \cos 74 = 13.165 \text{ lbf}$

$M_{z\,bend} = P_1 \times 2'' = 45.913 \text{ lbf} \times 2'' = 91.826 \text{ lbf-in}$

$M_{z\,Torsion} = P_2 \times 14.6'' = 13.165 \text{ lbf} \times 14.6'' = 192.209 \text{ lbf-in}$

find critical stress element

$I = \frac{\pi}{64} d^4 = \frac{\pi}{64}(1'')^4 = 0.0491 \text{ in}^4$

$J = \frac{\pi}{32} d^4 = \frac{\pi}{32}(1'')^4 = 0.0981 \text{ in}^4$

- Bending along z-axis $\rightarrow \sigma = \frac{Mc}{I} = \frac{91.826 \text{ lbf-in} \times 0.5\text{in}}{0.0491 \text{ in}^4} = 935.092 \text{ psi} \approx 935 \text{ psi}$

- Torsion around z-axis $\rightarrow \tau = \frac{Tr}{J} = \frac{192.209 \text{ lbf-in} \times 0.5\text{in}}{0.0981 \text{ in}^4} = 979.659 \text{ psi} \approx 980 \text{ psi}$

Factor of Safety using max shear stress

principle Stresses

$\sigma_1, \sigma_3 = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$ Plane stress $\rightarrow$ no σ_y

$\sigma_1 = \sigma + \sqrt{\left(\frac{\sigma}{2}\right)^2 + \tau_{xy}^2} = \frac{935}{2} + \sqrt{\left(\frac{935}{2}\right)^2 + 980^2} = 1553.298 \text{ psi}$

$\sigma_3 = \sigma - \sqrt{\left(\frac{\sigma}{2}\right)^2 + \tau_{xy}^2} = \frac{935}{2} - \sqrt{\left(\frac{935}{2}\right)^2 + 980^2} = -618.298 \text{ psi}$

Factor of safety

$FOS = \eta = \frac{S_y}{\sigma_1 - \sigma_3} = \frac{50800 \text{ psi}}{1553.298 - (-618.298) \text{ psi}} = \boxed{23.39}$

Maximum Drop Distance (failure)

find force to cause yielding

$$\sigma = \frac{Mc}{I} = \frac{F\sin(74)\times 2\text{in}\times 0.5\text{in}}{0.0491\ \text{in}^4} = 19.578F$$

$$\tau = \frac{Tr}{J} = \frac{F\cos(74)\times 14.6\text{ in}\times 0.5\text{ in}}{0.0981\ \text{in}^4} = 20.511F$$

Distortion Energy $\rightarrow$ $\sigma' = \sqrt{\sigma^2 + 3\tau_{xy}}$

$$\sigma' = \sqrt{(19.578F)^2 + 3(20.511)^2} = 40.564F$$

$$FoS = \frac{S_y}{\sigma'} \longrightarrow FoS = 1 \text{ for yield} \longrightarrow 1 = \frac{50800}{40.564F} \rightarrow F = 1252.342 \approx \boxed{1252\ \text{lbf}}$$

$\hookrightarrow$ this is for one blade only

Force to cause yielding is approximately 1252 lbF.

Impulse – momentum

$$m = \frac{W}{g} = \frac{200\ \text{lb-f}}{32.2\ \text{ft/s}^2} = 6.211\ \text{slugs} \quad \text{and } \Delta t = 0.05\ \text{sec}$$

$$F_{impact} = \frac{mv}{\Delta t} = \frac{m\sqrt{2(32.2)h}}{\Delta t} = \frac{6.211\sqrt{64.4h}}{0.05} = 124.2\times 8.025\sqrt{h} = 997\sqrt{h}$$

$$F_{yield} = \underbrace{2\times 1252\ \text{lbF}}_{\text{2 blades!}} = F_{impact} \longrightarrow \underset{\text{lbF}=\frac{\text{slug}\times\text{ft}}{\text{s}^2}}{2504} = \underset{\text{slugs}\times\text{ft}}{997\sqrt{h}} \rightarrow h = \boxed{6.35\text{ ft or } 76.2\text{ in}}$$

find radius to increase drop hieght by 50%

new hieght $= 6.35 + (6.35\times 0.50) = 9.53$ ft

$$F_{impact} = \frac{997\sqrt{9.53}}{2} = 1538.9 \approx 1539\ \text{lbF}$$ Divide by 2 for force on blade

stress in terms of radius

$$\sigma = \frac{1539\sin(74)\times 2\times r}{\frac{\pi}{4}r^4} = \frac{4\times 5917.527}{\pi r^3} = \frac{23670.108}{\pi r^3}$$

$$\tau_{xy} = \frac{1539\cos(74)\times 14.6\times r}{\frac{\pi}{2}r^4} = \frac{2\times 6193.406}{\pi r^3} = \frac{12386.812}{\pi r^3}$$

$$\sigma' = \left[\left(\frac{23670}{\pi r^3}\right)^2 + 3\left(\frac{12386.812}{\pi r^3}\right)^2\right]^{1/2} = \frac{60830.436}{\pi r^3}$$

$$Fos = 1 = \frac{\sigma_y}{\sigma'} \rightarrow \sigma' = \sigma_y$$

$$\frac{60830.436}{\pi r^3} = 50800\ psi \longrightarrow r = \sqrt{\frac{60830.436}{\pi \times 50800}} = 0.725\ in \longrightarrow D = 1.45\ in$$

optimal diameter = $\boxed{1.60\ in}$